E50 review pack: Fig. 4g and the text, after the STOP 3 rulings and their addendum

For the author. Exploratory and post hoc.

2026-11-03

What this is. Fig. 4g as E50 draws it, at final size (87.6 mm wide). With it come the text and the 4g legend as the author's STOP 3 rulings and their addendum fix them, the STYLE.md metrics, and the nine Extended Data panels the rulings put in the paper.

- **The rulings:** `docs/2026-11-03_stop3_rulings_consolidated.md` (`cb9b7ce`), committed verbatim and alone.
- **The addendum:** `docs/2026-11-03_stop3_rulings_addendum.md` (`3f9ffbe`), committed verbatim and alone.
- **The binding ruling:** `docs/2026-11-04_binding_classes_ruling.md` (`66c205a`), committed verbatim and alone.
- **Nothing new is computed.** Every number drawn was stored by E49 (`4fd661b`) and drawn by E50, which was verified (`docs/2026-11-03_e50_result.md`).

The text (rulings section 3, as the addendum amends it).

- **No sentence is Claude Code's.**
  - The sentences are the rulings' own.
  - R2 carries the search's cohorts, from the addendum.
  - The Methods definition is E47 declaration 4a.7, item 5, verbatim, with the addendum's two changes.
- **36 comparisons, made each time this pack is built, found no difference.** They check the text against the rulings, the addendum, E47′s declaration and the binding ruling.
- **The 4g legend is E50′s draft with the rulings' section 4 edits, items 1 to 6, and the binding ruling's item 7.**
- **It supersedes E49′s result note, Part 4, as committed** (`4fd661b`). The note now carries this text.

The addendum's answers to the four observations.

1. **R2 now gives the cohorts the search used,** n = 1,095 and 3,207. The model sets, 938 and 3,143, stay in the 4g legend, and the abstract's 4,081 stands.
2. **The brief's rule 5.4 now reads:** "BAD is reported with its explanation in the sentence that sets its rise against BCL-xL's, or the next." R7 meets it, and R4 stands unchanged.
3. **N4, N5 and N6 are dropped.** The Methods definition is E47 4a.7, item 5, verbatim, with two changes:
   - the construction sentence now says that selection was at fixed proliferation. It says the association with proliferation is in part expected by construction, and that the informative quantity is MYC activity's at fixed OXPHOS and proliferation (D8 and D9);
   - the variants sentence ends with "(Extended Data Fig. Xg)" (D10).

   N1 to N3 stay. N6, as ruled, misdescribed S5, whose rule uses MYC activity; E47′s paragraph states it correctly (D10, D11).
4. **ED-g keeps its ±0.9 scale.** The author deletes the "Proliferation minus OXPHOS" column in Illustrator. No legend edit and no new script follow.

The binding classes (the binding ruling, 2026-11-04; `data/annotations/bh3_binding_code.csv`).

- **BID binds both BCL-xL and MCL1.** BIK and BMF bind BCL-xL and, by the author's decision, are coded as not binding MCL1: they strongly favour BCL-xL, and their MCL1 binding is weak, not absent.
- **The sources are Ku et al. 2010 and Wang et al. 2023**, in place of Chen et al. 2005′s full data (rulings section 5). The legend (item 7) and Methods note N2 cite all three.
- **All seven rows are now filled, so legend edit 6 applies.** The code reaches 4g only through the author's Illustrator annotations. E49, which read the table with these rows blank, is not re-run.

For the author: the Methods definition, measured (addendum 3: any sentence outside the range is reported for the author to split).

- **4 sentences are outside STYLE.md's range, 12 to 38 words, all in: Methods definition.** Each is E47′s wording, kept verbatim, for the author to revise.
  - **Over 38, to split:** D3 (58 words) and D10 (43 words).
  - **Under STYLE.md's hard floor of 12:** D4 (8 words) and D12 (7 words). Splitting cannot fix these; joining can.
  - **The construction sentences, D8 and D9, are inside the range** (32 and 14 words).
- **D10 ends "(omitting BID) (Extended Data Fig. Xg)."**, as addendum 3b reads. The two parentheses now sit side by side.

The displays (rulings section 1; addendum 4).

- **Accepted as E50 drew them:**
  - the palette from Fig. 1g's source, interpolated in CIELAB;
  - one ±0.6 scale shared by 4g, ED-e and ED-f;
  - the 1.6 mm gap before the Signature row;
  - white ink on 9 cells;

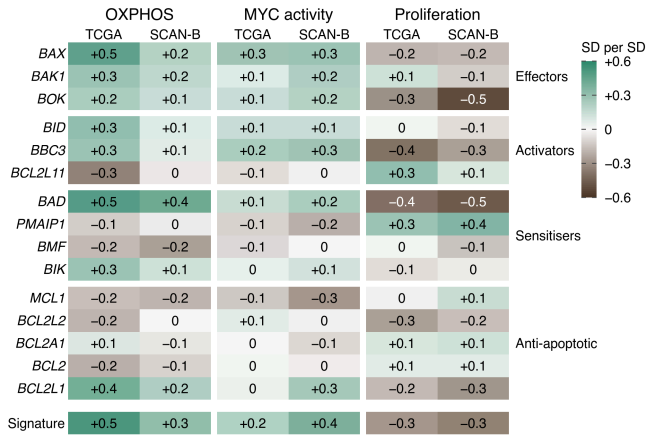
  - ED-g's ±0.9 scale.
- **The author's, in Illustrator, outside the repo's scripts.** No new display script is needed.
  - **Type sizes:** the group names and the colour bar's title at 6 pt, as in Fig. 1g.
  - **4g's annotations:** the binding code, the signature signs, the § marks and any protein names.
  - **ED-g:** the "Proliferation minus OXPHOS" column is deleted.
- **So the panels below are E50′s PNG previews, before that work.** The PDF and SVG beside each in `docs/figures/` are the files for Illustrator, with live Helvetica text.
- **Extended Data in the paper:** ED-a, ED-b, ED-e, ED-f, ED-g, ED-h, ED-i, ED-j and ED-k, each cited.
  - **In the repo, out of the paper:** ED-c (`docs/figures/2026-11-03_E48_myc_score_ladder`), superseded by ED-i and ED-j, and S-1 (`2026-11-03_E48_sensitivities_tp53_composition`), kept for referee responses.
  - **ED-d** stays superseded by ED-i.

H2 (rulings section 2).

- **Accepted.** The text says first that the signature was selected for tracking OXPHOS at fixed proliferation.
- **The signature's associations with OXPHOS and proliferation are what the selection intended.** The independent evidence is *BBC3* and *BCL2L1*, MYC activity, and the effectors.
- **The H2 test is not declared.** It is the answer if a referee asks how much of the signature's proliferation coefficient the selection explains.

How the metrics are counted. Words are whitespace tokens containing a letter or digit. Figure citations in parentheses and reference numbers are not counted.

Fig. 4g (E50)



E50's preview, before the author's Illustrator work: no annotations, and the group names and the bar's title at 5 pt, which become 6 pt.

The text: the rulings' section 3, as the addendum amends it

Sentence ids are for reference.

Abstract: the human sentences. [A1] Extending this relationship to 4,081 human breast tumours[8,9], a BCL-2 family signature that tracked OXPHOS at fixed proliferation showed BCL-xL rising more steeply than PUMA. **[A2]** MYC activity, which played no part in defining it, went with the same signature when measured with scores least correlated with proliferation.

Mouse Fig. 4 paragraph: last sentence. [ML] Consequently, established human tumours with high OXPHOS and MYC activity were predicted to carry raised pro-apoptotic signalling restrained by BCL-xL, a configuration distinct from that accompanying proliferation.

Results, paragraph 1: the selection. [R1] In tumours, however, OXPHOS rises with both MYC activity and proliferation (Extended Data Fig. Xa), so testing this prediction required OXPHOS to be separated from proliferation. **[R2]** We therefore searched the twelve BH3-only and anti-apoptotic genes of the BCL-2 family in TCGA-BRCA (n = 1,095) and SCAN-B (n = 3,207) for those whose expression tracked OXPHOS at fixed proliferation. **[R3]** Six of the eleven mitochondrial members did, and together they formed a signature (Fig. 4g). **[R4]** BCL2L1, encoding BCL-xL, rose with the activators BBC3 and BID and the sensitisers BAD and BIK, whilst MCL1 fell.

Results, paragraph 2: the balance. [R5] Its structure fitted the prediction, BCL2L1 rising more steeply with OXPHOS than BBC3, encoding PUMA, in both cohorts (Extended Data Fig. Xb). **[R6]** This is the reverse of the MYC-expressing gland, in which PUMA rose relative to BCL-xL with OXPHOS priority (Fig. 1i; Extended Data Fig. Xk), and resembles the mouse tumours that sustained high OXPHOS only alongside Bcl-xL (Fig. 3h,i). **[R7]** BAD rose more steeply than BCL2L1 (Fig. 4g), but BAD neutralises BCL-xL only when unphosphorylated, and survival signals phosphorylate it and hold it away from BCL-xL[Zha]. **[R8]** Its expression therefore does not measure its activity, unlike that of PUMA, which is controlled mainly through transcription[Nakano].

Results, paragraph 3: MCL1. [R9] MCL1 fell as BCL-xL rose, so BCL-xL was the only one of the two main guardians to rise alongside PUMA, which binds both[Chen]. **[R10]** MCL1's reported support of respiration in breast cancer stem cells depends on a mitochondrial function separable from its anti-apoptotic role[27]. **[R11]** This suggests that tumours with high OXPHOS rely on BCL-xL rather than MCL1, like persistent OXPHOS-high triple-negative breast cancer cells, which depend on BCL-xL and are insensitive to MCL1 inhibition[26].

Results, paragraph 4: the joint model. [R12] With OXPHOS, MYC activity and proliferation modelled together and intrinsic subtype adjusted for, the signature followed OXPHOS but not proliferation, as its selection intended (Fig. 4g). **[R13]** So did BBC3 and BCL2L1, the two genes named by the mouse before any selection (Extended Data Fig. Xh). **[R14]** MYC activity played no part in the selection, yet it went with the signature in the same direction as OXPHOS (Fig. 4g). **[R15]** This held on the two MYC scores least correlated with proliferation, and was weaker on scores that track proliferation closely (Extended Data Fig. Xa,j). **[R16]** The effectors BAX, BAK1 and BOK, outside the signature, also rose with OXPHOS and MYC activity in both cohorts (Fig. 4g).

The clinical paragraph: the new opening sentence, then its two unchanged sentences. [C1] Having separated OXPHOS from proliferation on PUMA and BCL-xL, we next asked whether the two are also separable in their association with chemotherapy response. **[C2]** Additional adjustment for MYC activity, and then for the mean expression of BCL2L1 and MCL1, did not materially alter the OXPHOS–response association (pooled odds ratios 0.87 and 0.88). **[C3]** The transcriptional coupling of OXPHOS and proliferation therefore concealed a clinically relevant distinction.

The closing paragraph: its human sentences. [Z1] In human breast tumours, BCL-xL rose more steeply than PUMA as OXPHOS rose, and MYC activity went with the OXPHOS-associated signature. **[Z2]** Proliferation, although correlated with OXPHOS, was not part of this configuration and went the other way on chemotherapy response.

Methods: the signature's definition (E47 declaration 4a.7, item 5, verbatim, with the addendum's two changes). *BCL-2-family genes and signature.* **[D1]** We examined twelve BCL-2-family genes that set the threshold for mitochondrial apoptosis: seven BH3-only proteins (BCL2L11/BIM, BID, BBC3/PUMA, BAD, BIK, BMF and PMAIP1/NOXA) and five anti-apoptotic proteins (BCL2, BCL2L1/BCL-xL, MCL1, BCL2L2/BCL-w and BCL2A1/BFL-1). **[D2]** For each gene we computed the Spearman correlation of its expression with the OXPHOS subunit score (GSVA), partial on a proliferation score, in TCGA-BRCA and SCAN-B, and averaged the two cohorts. **[D3]** The signature comprised every gene that is a mitochondrial protein (Human MitoCarta 3.0: eleven of the twelve, all outer-membrane; BMF is not listed) and whose mean correlation with OXPHOS was at least 0.20 in absolute value: BAD, BCL2L1, BIK, BBC3 and BID, which rise with

OXPHOS (mean rho 0.21 to 0.50), and MCL1, which falls (mean rho −0.27). **[D4]** Selection did not use correlation with MYC activity. **[D5]** BID qualified narrowly (0.21) and is the one member more strongly correlated with MYC activity (0.30) than with OXPHOS. **[D6]** Each gene was assigned a signature sign, +1 if it rises with OXPHOS and −1 if it falls, recording direction, not pro- or anti-apoptotic function. **[D7]** A tumour's signature score was the mean of the six signed z-scores (log2-normalised expression, standardised across the cohort), rank-normalised within each cohort. **[D8]** Because genes and signs were chosen by their association with OXPHOS at fixed proliferation in the same cohorts, the signature's association with OXPHOS, and in part with proliferation, is expected by construction. **[D9]** The informative quantity is its association with MYC activity at fixed OXPHOS and proliferation. **[D10]** All analyses were repeated with two rule-based variants: the same rule without the mitochondrial restriction (adding BMF, sign −1); and the five genes whose correlation with OXPHOS exceeded their correlation with MYC by at least 0.20 in at least one cohort (omitting BID) (Extended Data Fig. Xg). **[D11]** The second variant is selected against association with MYC and is reported for transparency, not as an independent test. **[D12]** All twelve genes are also reported individually.

Methods notes (E49′s three, kept). [N1] The effectors BAX, BAK1 and BOK were fitted for comparison with the groups of Fig. 1g and were not added to the signature. **[N2]** Binding of BCL-xL and MCL1 by each BH3-only protein was taken from published peptide affinities and is shown only where the source states it[Chen,Ku,Wang]. **[N3]** This analysis was post hoc, and the signature's proliferation coefficient had been seen in an earlier analysis before it was declared; no reading rule was applied, and every coefficient is reported.

The legend: E50′s draft, with the rulings' section 4 edits

261 words; the draft has 177. Associations of OXPHOS level, MYC activity and proliferation with the expression of fifteen BCL-2 family genes, in the groups of Fig. 1g, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each row is one linear model of the gene's expression on OXPHOS, MYC activity and proliferation, with intrinsic subtype, and with tumour purity and leukocyte fraction in TCGA-BRCA; each coefficient is in SD per SD with the other two held fixed, and every variable is rank-normalised within each cohort. Cells give the coefficient, rounded to one decimal; intervals are in Extended Data Fig. Xi. MYC activity is scored on MYC_UP.V1_UP, one of the two MYC activity scores least correlated with proliferation; the reference score is used in Extended Data Fig. Xf, and all four in Extended Data Fig. Xj. Left, binding of BCL-xL and MCL1 by BH3-only proteins, from Chen et al. (2005), Ku et al. (2010) and Wang et al. (2023): filled, binds; open, does not. Signature sign: the direction in which each of the six signature genes enters the signature. § BH3-only proteins whose activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). Bottom, the six-gene signature, selected for its association with OXPHOS at fixed proliferation (Methods). OXPHOS priority is shown in Extended Data Fig. Xe. Outside the signature, BCL2L11, BMF and BCL2 also fell with OXPHOS. At fixed OXPHOS and MYC activity, proliferation went with lower BAX and BOK and higher PMAIP1 and BCL2L11 in both cohorts; its association with BAK1 differed in sign between cohorts, and MCL1 was higher only in SCAN-B.

item	the edit	status
1	adds the sentence on cells and intervals, after the modelling sentence	applied
2	the MYC activity sentence names its place among the four scores, and cites ED-f and ED-j	applied
3	the signature sentence names the bottom row ("Bottom, the six-gene signature")	applied
4	cites ED-e	applied
5	adds the two sentences on patterns outside the text, replacing E49′s proposed additions. The rulings do not place them; they are appended, where those additions stood	applied
6	deletes "; blank, not yet verified" once BID, BIK and BMF are entered	applied
7	cites Ku et al. (2010) and Wang et al. (2023) beside Chen et al. (2005), the sources of BID, BIK and BMF (the binding ruling)	applied

The model sets, n = 938 and 3,143, stay in this legend (addendum 1). STYLE.md does not yet cover legends (its section 9), so only the word count is given.

STYLE.md metrics

paragraph	sentences	words	mean	median	range	under 20	under 12	over 38	em-dashes
Abstract	2	47	23.5	23.5	22–25	0 of 2	0	0	0
Mouse Fig. 4, last sentence	1	27	27.0	27	27–27	0 of 1	0	0	0
Results 1, selection	4	84	21.0	20.5	13–30	2 of 4	0	0	0
Results 2, balance	4	90	22.5	21	18–30	2 of 4	0	0	0
Results 3, MCL1	3	73	24.3	23	20–30	0 of 3	0	0	0
Results 4, joint model	5	98	19.6	20	15–24	2 of 5	0	0	0
Clinical paragraph	3	65	21.7	24	13–28	1 of 3	0	0	0
Closing, human sentences	2	40	20.0	20	19–21	1 of 2	0	0	0

paragraph	sentences	words	mean	median	range	under 20	under 12	over 38	em-dashes
Methods definition	12	311	25.9	23.5	7–58	5 of 12	2	2	0
Methods notes	3	78	26.0	24	23–31	0 of 3	0	0	0

The four Results paragraphs together: 16 sentences, 345 words, mean 21.6, median 20, range 13–30, 6 of 16 (38%) under 20 words. These are the rulings' section 6 figures, reproduced. The addendum changes no word count there.

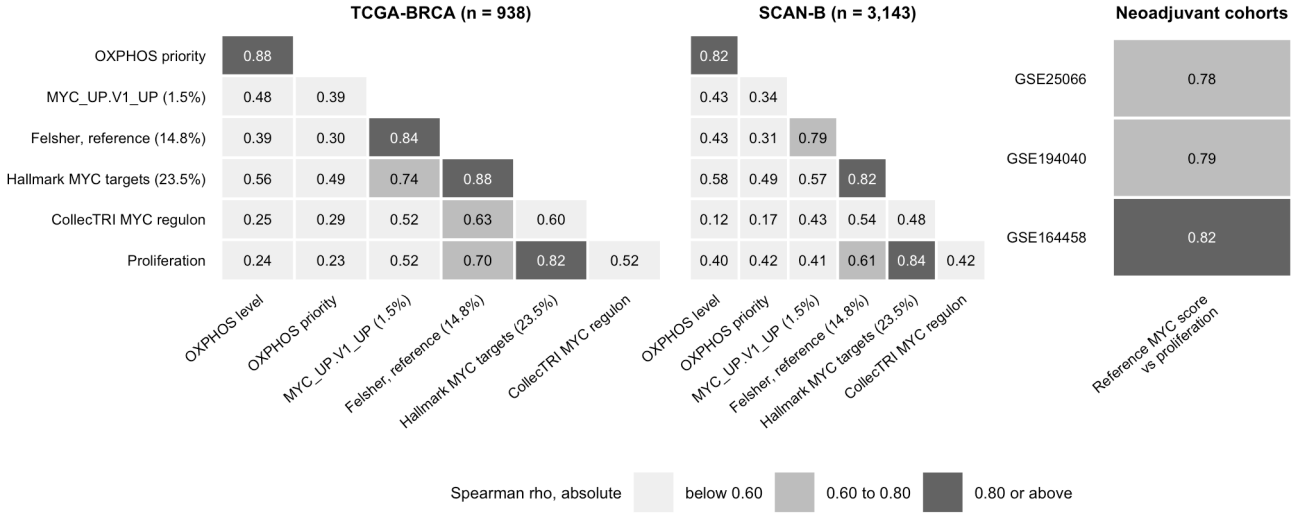
- **Against STYLE.md's targets:** the range and the share under 20 are inside them (12 to 38; 35% to 45%). The mean and the median sit just below them (22 to 25; 21 to 24).
- **4 sentences are outside STYLE.md's range, 12 to 38 words, all in: Methods definition.** Each is E47′s wording, kept verbatim, for the author to revise.
  - **Over 38, to split:** D3 (58 words) and D10 (43 words).
  - **Under STYLE.md's hard floor of 12:** D4 (8 words) and D12 (7 words). Splitting cannot fix these; joining can.
  - **The construction sentences, D8 and D9, are inside the range** (32 and 14 words).
- **There are no em-dashes and no serial commas.** Each ", and" that the counter flags joins two clauses, two predicates or two groups, or opens a parenthesis.

The Extended Data panels in the paper

Each is shown at final size, with its legend draft verbatim from the script that drew it. The letters are placeholders: a and b are E48′s; e to h are E50′s, which redraws E49′s; i to k are E49′s. Where each is cited is read from the text and the legend above, and checked against the rulings' 1.4.

ED-a (E48): the scores' correlations

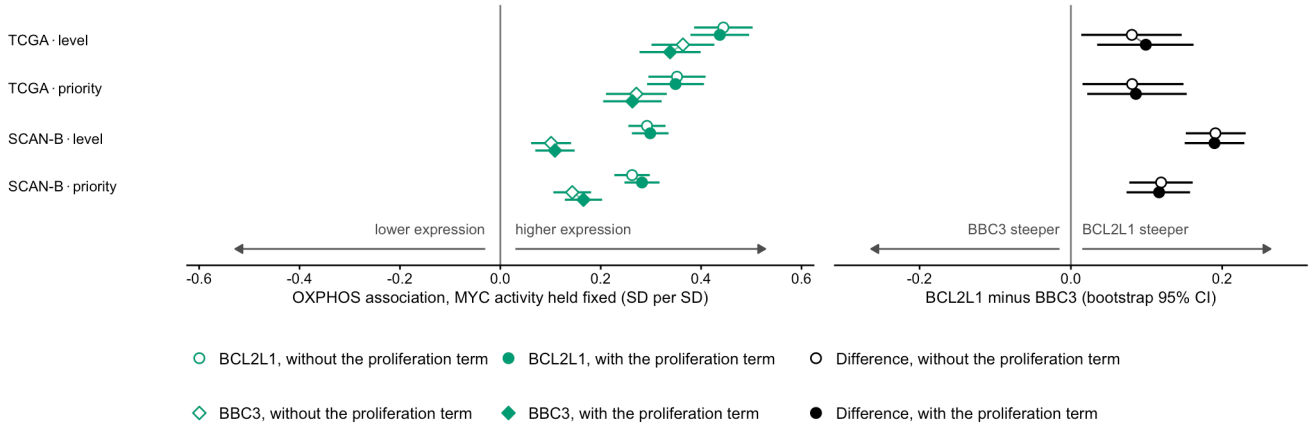
Cited as: Extended Data Fig. Xa, in sentences R1 and R15.



Legend draft (66 words). Spearman correlations among the two OXPHOS scores, four MYC activity scores and proliferation in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), shaded by band (below 0.60; 0.60 to 0.80; 0.80 or above). Each MYC activity score is labelled with the percentage of its genes in proliferation gene sets (Hallmark E2F targets and G2M checkpoint). Right, the reference MYC activity score against proliferation in three neoadjuvant cohorts.

ED-b (E48): *BCL2L1* against *BBC3*

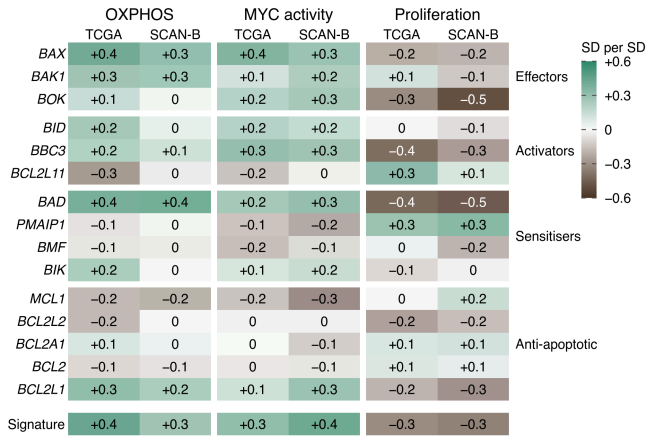
Cited as: Extended Data Fig. Xb, in sentence R5.



Legend draft (51 words). OXPHOS coefficients for BCL2L1 and BBC3 with MYC activity held fixed, without (open) and with (filled) the proliferation term, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143), and the difference between them, with 95% percentile intervals from 2,000 paired bootstrap resamples of tumours. Bars on the coefficients, model-based 95% confidence intervals.

ED-e (E50): Fig. 4g on OXPHOS priority

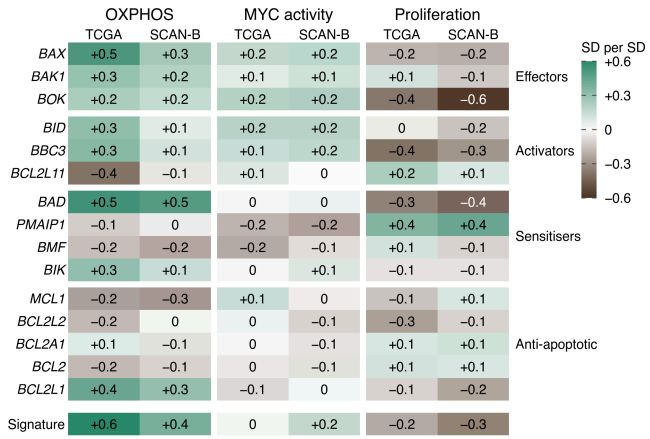
Cited as: Extended Data Fig. Xe, in the 4g legend.



Legend draft (14 words). As Fig. 4g, with OXPHOS scored as its priority within the mitochondrial programme (mitoPPS).

ED-f (E50): Fig. 4g from the reference model

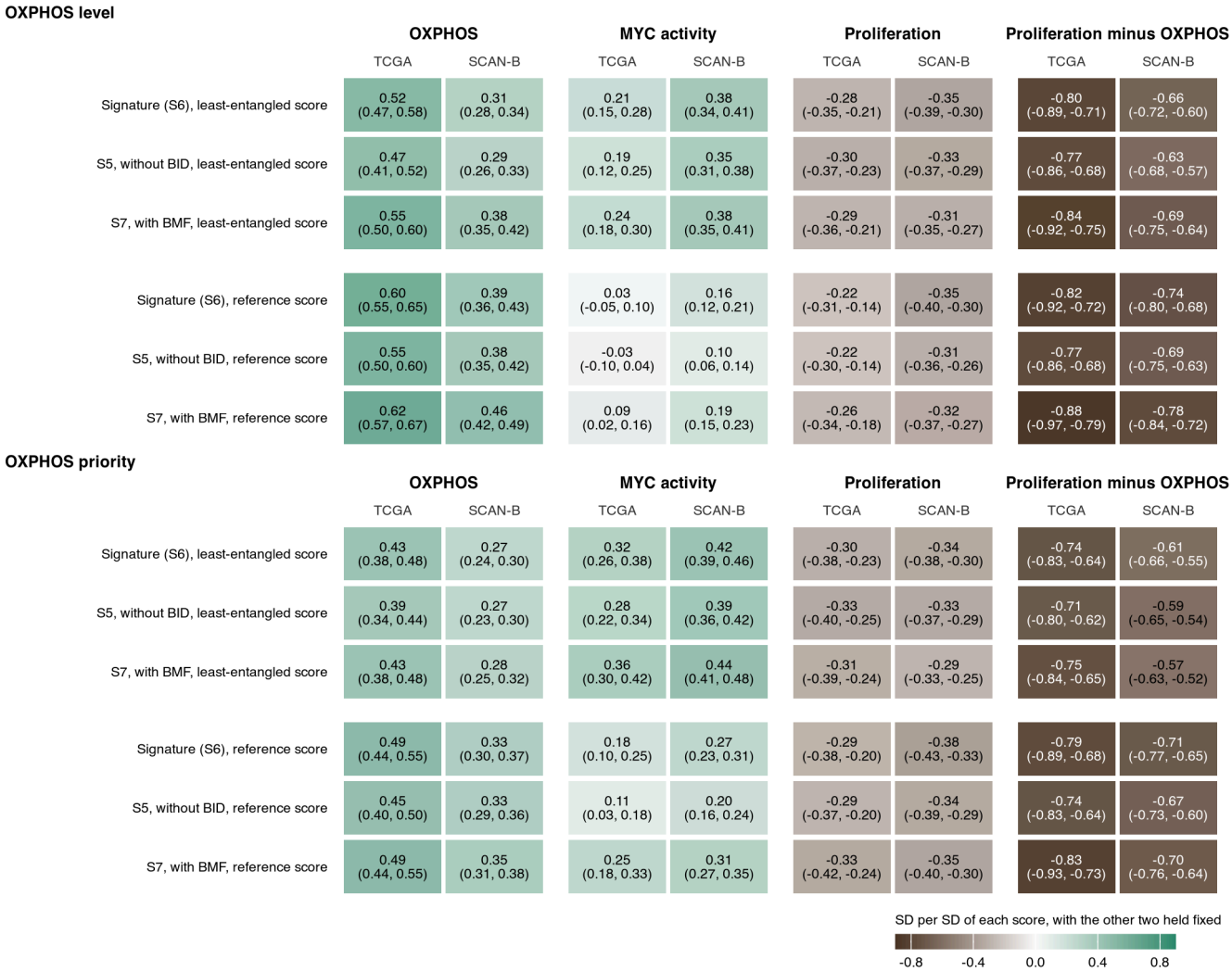
Cited as: Extended Data Fig. Xf, in the 4g legend.



Legend draft (20 words). As Fig. 4g, from models using the reference MYC activity score (Felsher), which correlates with proliferation at 0.61 to 0.70.

ED-g (E50): the signature and its two variants

Cited as: Extended Data Fig. Xg, in the Methods (D10).

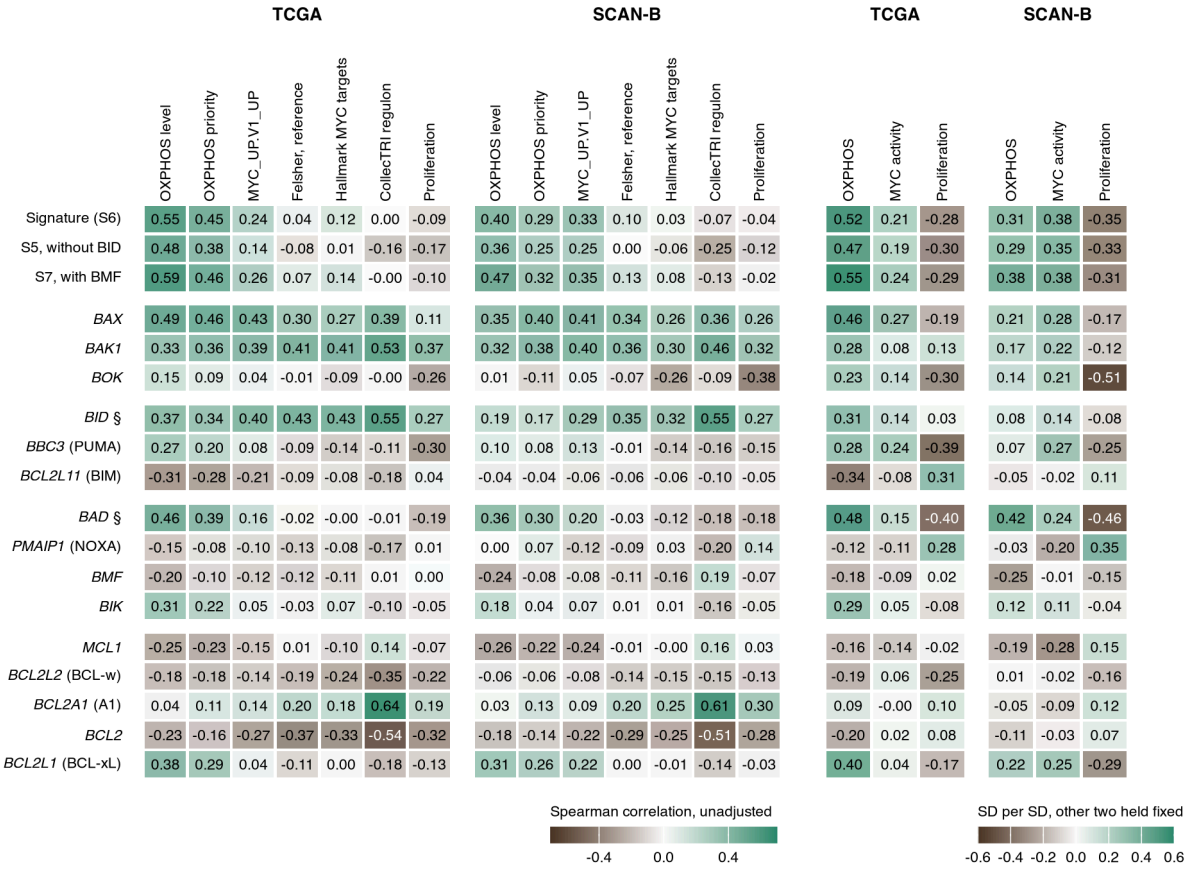


The author deletes the "Proliferation minus OXPHOS" column in Illustrator (rulings 1.3; addendum 4). It contains the coefficient that defines the signature, and it remains in E49′s tables. **ED-g keeps its ±0.9 scale.** No legend edit and no new script follow.

Legend draft (71 words). The six-gene signature (S6) and its two variants, S5 (without BID) and S7 (with BMF), on the MYC activity score with few proliferation genes and on the reference score, with OXPHOS scored as level and as priority, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Each cell gives the coefficient (SD per SD) and its 95% confidence interval. The signature was selected for its association with OXPHOS at fixed proliferation (Methods).

ED-h (E50): raw correlations beside the adjusted coefficients

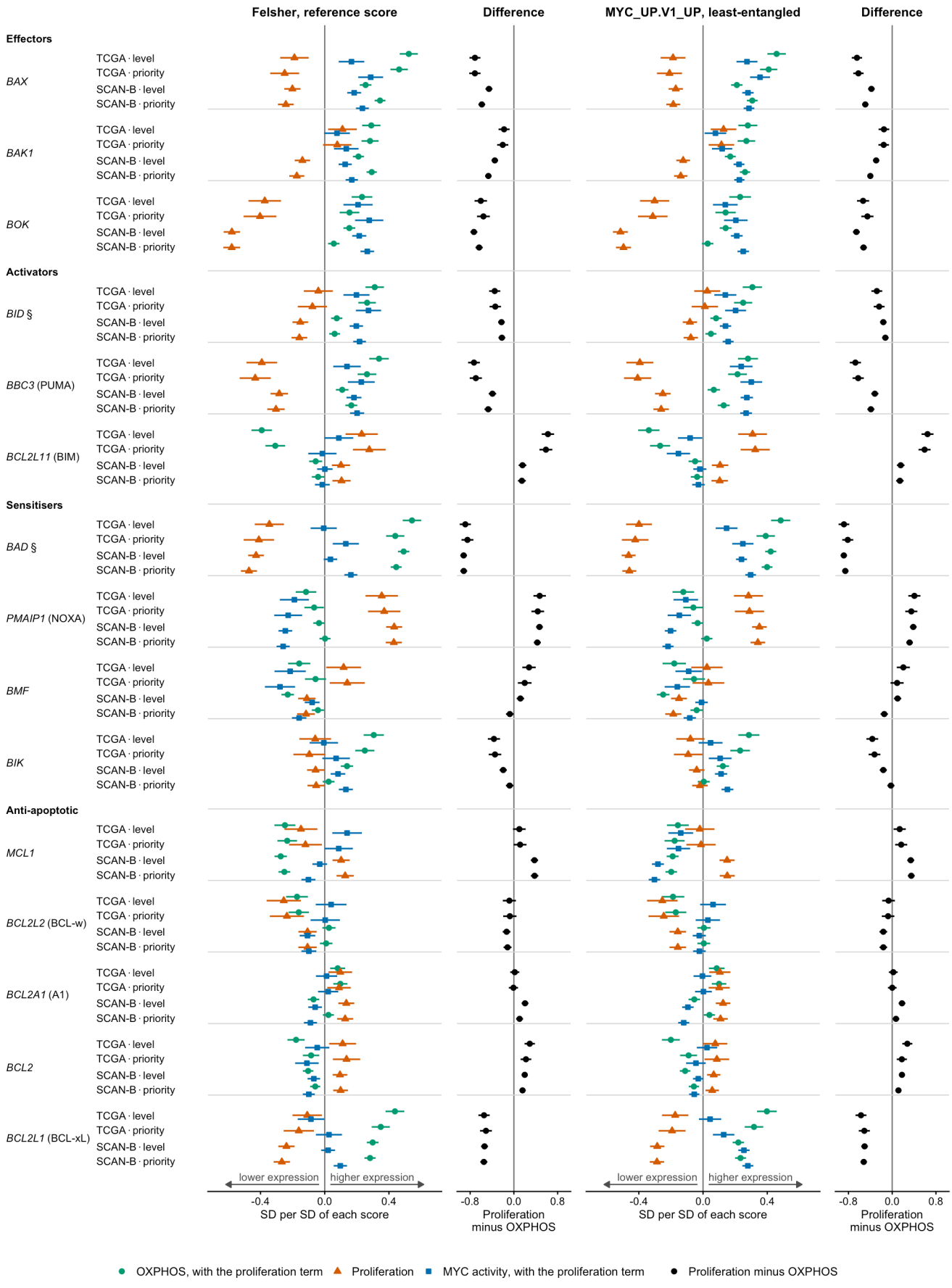
Cited as: Extended Data Fig. Xh, in sentence R13.



Legend draft (53 words). Left, unadjusted Spearman correlations of the signature, its variants and fifteen BCL-2 family genes with OXPHOS (level and priority), four MYC activity scores and proliferation, in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Right, the adjusted coefficients of Fig. 4g. The signature was selected for its association with OXPHOS at fixed proliferation (Methods).

ED-i (E49): all fifteen genes, three coefficients and the difference

Cited as: Extended Data Fig. Xi, in the 4g legend.

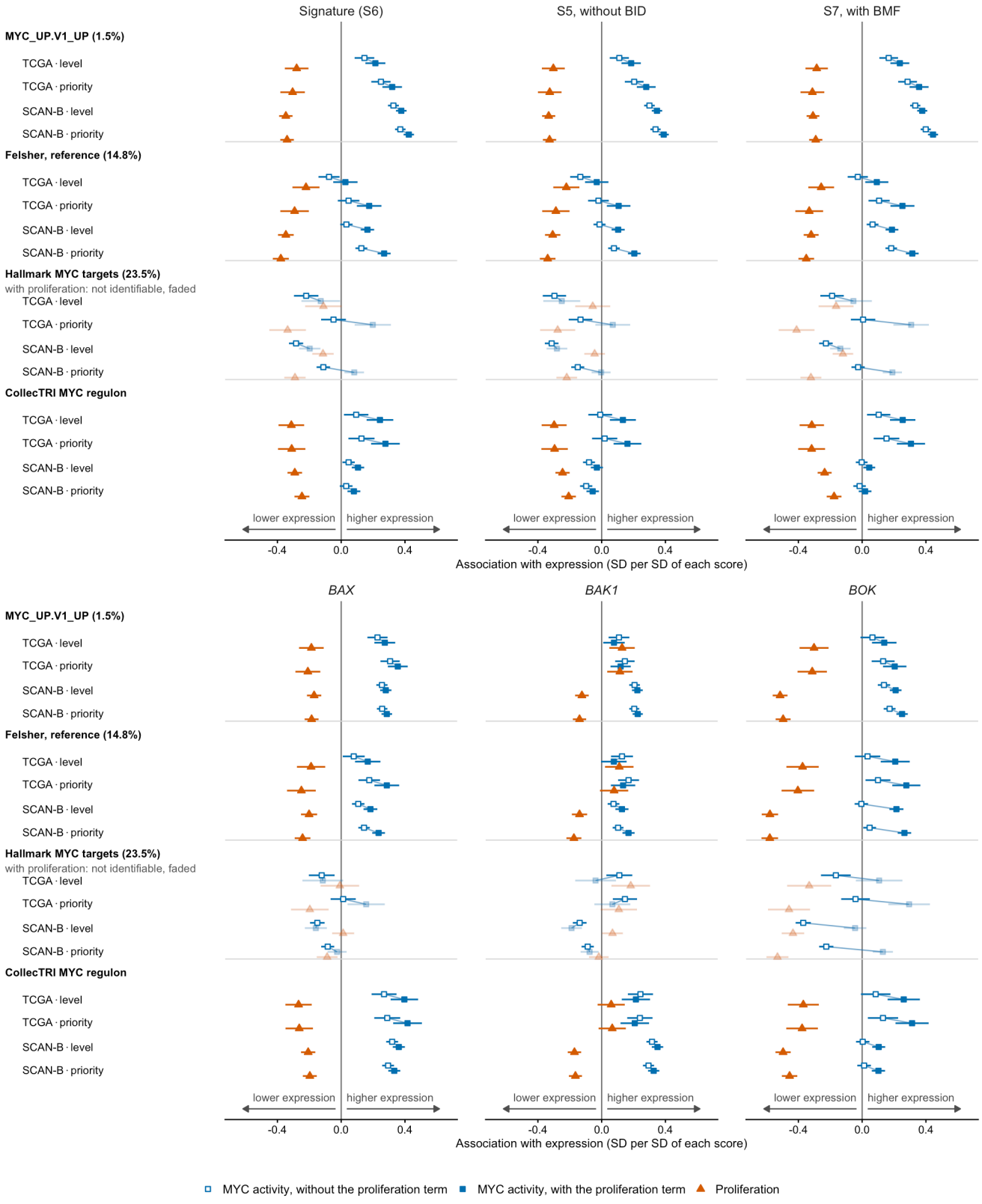


Legend draft (81 words). Fifteen BCL-2 family genes, in the groups of Fig. 1g: coefficients of OXPHOS (circles), proliferation (triangles) and MYC activity (squares), each with the other two held fixed, and proliferation minus OXPHOS, on the reference MYC activity score (left) and the score with few proliferation genes (right), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). § BH3-only proteins whose

activity is switched after translation (BAD: off by phosphorylation; BID: on by cleavage to tBID). No correction for multiple genes. Bars, 95% confidence intervals.

ED-j (E49): MYC activity on four scores, the signatures and the effectors

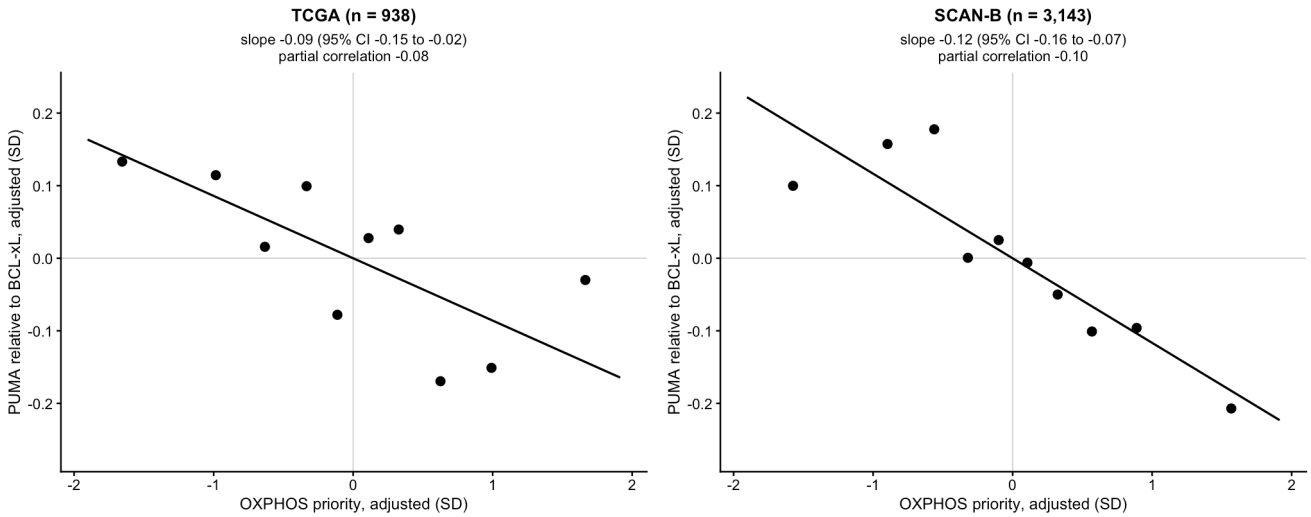
Cited as: Extended Data Fig. Xj, in sentence R15, and in the 4g legend.



Legend draft (76 words). MYC activity coefficients for the signature, its two variants and BAX, BAK1 and BOK on four MYC activity scores, ordered by the percentage of their genes in proliferation gene sets, without (open) and with (filled) the proliferation term, with the proliferation coefficient (triangles), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Faded points are cells in which the score and proliferation correlate at 0.80 or more; they are shown and not interpreted. Bars, 95% confidence intervals.

ED-k (E49): PUMA relative to BCL-xL against OXPHOS priority

Cited as: Extended Data Fig. Xk, in sentence R6.



Legend draft (61 words). BBC3 (PUMA) relative to BCL2L1 (BCL-xL) against OXPHOS priority, each adjusted for the reference MYC activity score, proliferation and intrinsic subtype (and purity and leukocyte fraction in TCGA-BRCA), in TCGA-BRCA (n = 938) and SCAN-B (n = 3,143). Points, means of the ten deciles of adjusted OXPHOS priority; line, the fitted slope, with its 95% interval from 2,000 paired bootstrap resamples of tumours.